THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Mathematics | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(b)** 30°	$\sin\theta = \frac{1}{2}$
2	**(d)** $\frac{4}{5}$	Sides 4, 3, 5 ⇒ $\sin A = \frac{4}{5}$
3	**(d)** 1	$\frac{1}{2} + \frac{1}{2}$
4	**(a)** 50 m	$100\sin 30^\circ$
5	**(b)** $100\sqrt{3}$ m	$100\tan 60^\circ$
6	**(d)** 5 m	$\frac{2.5}{\cos 60^\circ}$
7	**(b)** Student A is correct	$OP \perp PQ \Rightarrow \sqrt{169-25} = 12$
8	**(d)** point of contact	Definition
9	**(a)** 3 cm	$AB + CD = BC + AD \Rightarrow 10 = 7 + AD$
10	**(c)** $\frac{22}{3}$ cm	$\frac{1}{12} \times 2 \times \frac{22}{7} \times 14$
11	**(b)** $\frac{77}{8}$ cm²	$r = 3.5$; $\frac{1}{4} \times \frac{22}{7} \times 12.25$
12	**(c)** 77 cm²	$\frac{1}{2} \times 154$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$\frac{\frac{5}{4}+\frac{16}{3}-1}{1} = \frac{15+64-12}{12} = \frac{67}{12}$	1+1
14	Base 12, hyp 13, perp 5: $\sin = \frac{5}{13}, \cos = \frac{12}{13}, \tan = \frac{5}{12}, \cot = \frac{12}{5}, \csc = \frac{13}{5}$	2
15	Height = 3.7 m; $l = \frac{3.7}{\sin 60^\circ} = \frac{7.4}{\sqrt{3}} \approx 4.27$ m	1+1
16	Half chord $= \sqrt{100-36} = 8 \Rightarrow$ chord = 16 cm	1+1
17	$\frac{1}{8} \times \frac{22}{7} \times 2025 = \frac{22275}{28} \approx 795.54$ cm²	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$\tan A = \frac{3}{4}$: LHS $= \frac{7/16}{25/16} = \frac{7}{25}$; $\cos A = \frac{4}{5}$, $\sin A = \frac{3}{5}$: RHS $= \frac{16-9}{25} = \frac{7}{25}$ ✓ Yes	1+1+1
19	$h\sqrt{3} - \frac{h}{\sqrt{3}} = 40 \Rightarrow \frac{2h}{\sqrt{3}} = 40 \Rightarrow h = 20\sqrt{3} \approx 34.64$ m	1+1+1
20	Joining O to the vertices and points of contact gives 8 angles in 4 equal pairs; $2(a+b+c+d) = 360^\circ \Rightarrow \angle AOB + \angle COD = a+b+c+d = 180^\circ$	1+1+1
21	$\frac{80}{360} \times 3.14 \times 16.5^2 = \frac{2}{9} \times 854.865 \approx 189.97$ km²	1+1+1
22	LHS $= \frac{(1-\cos\theta)^2}{\sin^2\theta} = \frac{(1-\cos\theta)^2}{(1-\cos\theta)(1+\cos\theta)} = \frac{1-\cos\theta}{1+\cos\theta}$	1+1+1
	OR: LHS $= \cos A + 1$; RHS $= \frac{(1-\cos A)(1+\cos A)}{1-\cos A} = 1 + \cos A$ ✓	1+1+1
23	$TP = TQ \Rightarrow \angle TPQ = \angle TQP = 90^\circ - \frac{1}{2}\angle PTQ$. $\angle OPQ = 90^\circ - \angle TPQ = \frac{1}{2}\angle PTQ \Rightarrow \angle PTQ = 2\angle OPQ$	1+1+1
	OR: $AB + CD = AD + BC$ (tangent property); $AB = CD$, $AD = BC \Rightarrow 2AB = 2AD \Rightarrow$ all sides equal ⇒ rhombus	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	$d = \frac{10}{\tan 30^\circ} = 10\sqrt{3} \approx 17.32$ m; height above eye $= d\tan 60^\circ = 30$ m ⇒ hill = 40 m	2+2+1
	OR: (i) $\frac{60}{\sqrt{3}} = 20\sqrt{3} \approx 34.64$ m (ii) $60 - 20\sqrt{3} \approx 25.36$ m (iii) False — top is seen below the horizontal, so the tower is shorter	2+2+1
25	(i) $\frac{1}{3} \times 88 \approx 29.33$ cm (ii) $\frac{616}{3} \approx 205.33$ cm² (iii) triangle $= \frac{1}{2}r^2\sin 120^\circ = 49\sqrt{3} = 84.77 \Rightarrow 120.56$ cm² (iv) $616 - 120.56 = 495.44$ cm²	1+1+2+1
	OR: $22 = \frac{\theta}{360} \times 88 \Rightarrow \theta = 90^\circ$ (1). Sector = 154, triangle = 98 ⇒ minor segment = 56 cm² (3); major = $616 - 56 = 560$ cm² (1)	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME — SET B

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Mathematics | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** 3	$1+1+1$
2	**(a)** 1	$\sec^2 A \cos^2 A = 1$
3	**(b)** Student B is correct	$\sin 2A \neq 2\sin A$ in general
4	**(b)** Student A is correct	$\tan\theta = \frac{1}{\sqrt{3}}$
5	**(c)** 45°	$\tan\theta = 1$
6	**(b)** decreases	$\tan\theta = \frac{h}{\text{shadow}}$
7	**(a)** 40°	$180^\circ - 140^\circ$
8	**(d)** Incorrect — no tangent can be drawn from an interior point	Every line through an interior point is a secant
9	**(b)** 15 cm	$\sqrt{289-64} = 15$
10	**(c)** Incorrect — that is the arc length; area $= \frac{\theta}{360^\circ}\pi r^2$	Formula mix-up
11	**(d)** $2r + \frac{\theta}{360^\circ} \times 2\pi r$	Two radii + arc
12	**(d)** 38.5 cm²	$r = 7$; $\frac{154}{4}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$A+B = 90^\circ,\ A-B = 30^\circ \Rightarrow A = 60^\circ,\ B = 30^\circ$	1+1
14	$\sin 30^\circ = \frac{1.5}{l} \Rightarrow l = 3$ m	1+1
15	Tangents at A and B are $\perp$ to diameter AB (Th. 10.1). Two lines perpendicular to the same line make equal alternate angles (90°) ⇒ parallel.	1+1
16	$\frac{30}{360} \times 3.14 \times 16 = 4.19$ cm² (approx.)	1+1
17	$11 = \frac{\theta}{360} \times 44 \Rightarrow \theta = 90^\circ$	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$AC = 25$. (i) $\sin A = \frac{7}{25},\ \cos A = \frac{24}{25}$ (ii) $\sin C = \frac{24}{25},\ \cos C = \frac{7}{25}$	1+1+1
19	Height above eye $= 28.5$ m. Distance $= 28.5\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = \frac{57}{\sqrt{3}} = 19\sqrt{3} \approx 32.9$ m	1+1+1
20	OP $\perp$ AB (radius $\perp$ tangent). The perpendicular from the centre to a chord bisects the chord ⇒ $AP = PB$	1+1+1
21	Sector $= \frac{1}{3} \times 3.14 \times 144 = 150.72$; triangle $= \frac{1}{2}r^2 \sin 120^\circ = 36\sqrt{3} = 62.28$; segment $= 88.44$ cm²	1+1+1
22	$h = x\sqrt{3} = \frac{x+20}{\sqrt{3}} \Rightarrow x = 10$; width 10 m, height $10\sqrt{3} \approx 17.32$ m	1+1+1
	OR: Let height H, distance $d = H$. $H - 8 = \frac{H}{\sqrt{3}} \Rightarrow H = \frac{8\sqrt{3}}{\sqrt{3}-1} = 4(3+\sqrt{3}) \approx 18.93$ m; distance ≈ 18.93 m	1+1+1
23	Arc $= 16.4 - 10.4 = 6$ cm; area $= \frac{1}{2} \times 6 \times 5.2 = 15.6$ cm²	1+1+1
	OR: $\frac{1}{8} \times 2464 = 308$ m²; cost = ₹1232	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) 13 cm (ii) $\sin = \frac{5}{13}, \cos = \frac{12}{13}, \tan = \frac{5}{12}, \csc = \frac{13}{5}, \sec = \frac{13}{12}, \cot = \frac{12}{5}$ (iii) $1 + \frac{25}{144} = \frac{169}{144}$ ✓	1+3+1
	OR: $p^2 - 1 = 2\sin\theta\cos\theta$; $q = \frac{\sin\theta+\cos\theta}{\sin\theta\cos\theta} = \frac{p}{\sin\theta\cos\theta} \Rightarrow q(p^2-1) = 2p$	2+2+1
25	Let $AF = AE = x$: $AB = x+8,\ AC = x+6,\ BC = 14,\ s = x+14$. Area $= r \cdot s = 4(x+14) = \sqrt{(x+14)\cdot x \cdot 6 \cdot 8}$ ⇒ $x + 14 = 3x \Rightarrow x = 7 \Rightarrow AB = 15$ cm, $AC = 13$ cm	2+2+1
	OR: Take any point Q on tangent XY other than P; Q lies outside the circle ⇒ $OQ > OP$. So OP is the shortest distance from O to XY ⇒ $OP \perp XY$ (3). Tangent $= \sqrt{225-81} = 12$ cm (2)	3+2

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ANSWER KEY & MARKING SCHEME — SET C

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Mathematics | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** $\sqrt{3}$	$\frac{2/\sqrt{3}}{2/3} = \sqrt{3} = \tan 60^\circ$
2	**(a)** $\frac{12}{13}$	$\sqrt{1 - \frac{25}{169}}$
3	**(b)** 0	$\frac{1-1}{1+1} = 0$
4	**(b)** $40\sqrt{3}$ m	$\frac{60}{\sin 60^\circ} = \frac{120}{\sqrt{3}}$
5	**(a)** 60°	$\tan\theta = \frac{6}{2\sqrt{3}} = \sqrt{3}$
6	**(b)** Student B is correct	$28.5 + 1.5 = 30$ m
7	**(b)** 8 cm	$\sqrt{100 - 36} = 8$
8	**(a)** 3 cm	$\sqrt{25 - 16}$
9	**(a)** Student A is correct	$AB + CD = BC + AD \Rightarrow AD = 6$
10	**(a)** 77 cm²	$\frac{1}{8} \times \frac{22}{7} \times 196 = 77$
11	**(b)** Incorrect — the area becomes 4 times	Area $\propto r^2$
12	**(a)** 22 cm	$\frac{1}{8} \times 176$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	LHS $= \sin 60^\circ = \frac{\sqrt{3}}{2} \approx 0.87$; RHS $= \frac{1}{2} + \frac{1}{2} = 1$. Not equal ⇒ claim is **false**	1+1
14	$\tan\theta = \frac{20}{20} = 1 \Rightarrow \theta = 45^\circ$. Claim is **false**.	1+1
15	$\angle OAT = 90^\circ \Rightarrow \angle OAB = 30^\circ = \angle OBA$ ($OA = OB$) ⇒ $\angle AOB = 120^\circ$	1+1
16	$\angle OAP = \angle OBP = \angle APB = 90^\circ$ and $OA = OB$ ⇒ OAPB is a square ⇒ $PA = 5$ cm. ($5\sqrt{2}$ is OP.) Claim is false.	1+1
17	Angle 30°: $\frac{1}{12} \times \frac{22}{7} \times 196 = \frac{154}{3} \approx 51.33$ cm²	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$PR^2 - QR^2 = 25 \Rightarrow PR - QR = 1 \Rightarrow PR = 13,\ QR = 12$. $\sin P = \frac{12}{13},\ \cos P = \frac{5}{13},\ \tan P = \frac{12}{5}$	1+1+1
19	Pedestal h, distance h (45°). $\sqrt{3} = \frac{h+1.6}{h} \Rightarrow h = \frac{1.6}{\sqrt{3}-1} = 0.8(\sqrt{3}+1) \approx 2.19$ m	1+1+1
20	$PA = PB$, $\angle P = 60^\circ$ ⇒ equilateral ⇒ $AB = 12$ cm; $\angle PAB = 60^\circ \Rightarrow \angle OAB = 90^\circ - 60^\circ = 30^\circ$	1½+1½
21	(i) $\frac{1}{5} \times 220 = 44$ cm (ii) $\frac{1}{5} \times 3850 = 770$ cm² (iii) $44 + 70 = 114$ cm	1+1+1
22	Given, To prove, figure. Any point Q ($\neq P$) on the tangent is outside the circle ⇒ $OQ > OP$ ⇒ OP is the shortest distance ⇒ $OP \perp$ tangent	1+2
	OR: Tangents at P and Q meet at T. $TP = TQ$ ⇒ $\triangle TPQ$ is isosceles ⇒ $\angle TPQ = \angle TQP$	1+1+1
23	$\frac{\sec^2 A}{\csc^2 A} = \tan^2 A$; $\frac{1-\tan A}{1-\frac{1}{\tan A}} = \frac{(1-\tan A)\tan A}{\tan A - 1} = -\tan A$, squared $= \tan^2 A$	1½+1½
	OR: $\sin A = \frac{1}{\sqrt{1+\cot^2 A}}$, $\sec A = \frac{\sqrt{1+\cot^2 A}}{\cot A}$, $\tan A = \frac{1}{\cot A}$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(ii) $\tan 45^\circ = \frac{10}{d} \Rightarrow d = 10$ m (iii) part above window $= 10\tan 60^\circ = 10\sqrt{3}$; height $= 10(1+\sqrt{3}) \approx 27.32$ m	1+2+2
	OR: (ii) $80\sqrt{3} \approx 138.56$ m and $\frac{80}{\sqrt{3}} \approx 46.19$ m (iii) $\frac{320}{\sqrt{3}} = \frac{320\sqrt{3}}{3} \approx 184.75$ m (iv) The one seeing it at 60° — larger angle, smaller distance	1+2+1+1
25	(i) 60° (ii) $\frac{1}{6} \times 1386 = 231$ cm² (iii) $\frac{1}{6} \times 132 = 22$ cm (iv) $2 \times 231 = 462$ ✓	1+2+1+1
	OR: (i) $\frac{3}{8} \times 2464 = 924$ cm² (ii) $\frac{3}{8} \times 176 = 66$ cm (iii) $132 + 462 = ₹594$	2+1+2

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